

PARTS LIST

Semiconductors:

IC1	- HT12E encoder
IC2	- 7805, 5V regulator
IC3	- HT12D decoder
T1-T4	- BC547 npn transistor
LED1-LED5	- 5mm LED
D1-D4	- 1N4148 signal diode
D5-D8	- 1N4007 rectifier diode
Board1	- Arduino Uno board
TX1	- 434MHz RF transmitter module
RX1	- 434MHz RF receiver module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 820-kilo-ohm
R2, R7-R10	- 1-kilo-ohm
R3-R6	- 10-kilo-ohm
R11	- 47-kilo-ohm

Capacitors:

C1	- 10 μ F, 16V electrolytic
C2	- 10 μ F, 25V electrolytic

Miscellaneous:

CON1	- 4-pin berg strip female connector for TX1
CON2	- 8-pin berg strip male connector
CON3	- 2-pin terminal connector
CON4-CON7	- 3-pin female connector
CON8	- 8-pin berg strip female connector for RX1
ANT.1, ANT.2	- 17cm single-strand wire antenna
RL1-RL4	- 9V, 1C/O relay
	- USB cable
	- 9V PP3 battery

When the relay is not energised, N/C and C terminals are short-circuited, and N/O and C terminals open-circuited. When the relay coil is energised, N/C and C terminals are open-circuited, and N/O and C terminals short-circuited.

Circuit diagrams of transmitter and receiver units are shown in Figs 5 and 6, respectively.

When a pushbutton is pressed in the GUI, the code executes a callback function in MATLAB. Commands to set or reset digital I/O pins 9 through 12 of the Arduino are written within that 'callback' function. After a pin is set or reset, the encoder scans the status of digital pins and transmits it wirelessly to the receiver side using the 434MHz wireless transmitter.

The transmitted data is received by the receiver and decoded to energise or de-energise the relays accordingly. When a particular digital I/O pin is set high by pressing the pushbutton, the corresponding relay

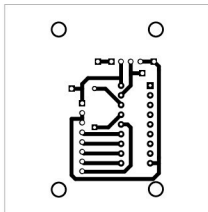


Fig. 7: Actual-size PCB layout of the transmitter circuit

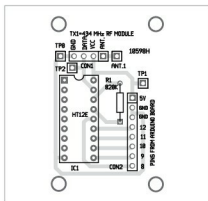


Fig. 8: Components layout for the PCB in Fig. 7

energises. Four BC547 BJTs (T1 through T4) are additionally used, as IC3 cannot directly source sufficient current to energise relays.

Diodes D1 through D4 are used as protection diodes to prevent sinking of current into IC3. Diodes D5 through D8 are fly-back diodes, which prevent damage to electronic components due to the back emf produced by the inductive action of the relay's coil. LED1 indicates power-on status of the circuit, whereas LED2 through LED5 show the status of the four relays. The power to the relay board is provided separately from a +9V battery.

Software program

We have used Arduino IDE for programming the Arduino Uno. The latest IDE can be downloaded for free from Arduino's official website. After downloading the Arduino

IDE, install it. Note the directory of the Arduino IDE installation. Follow the steps given below for this project:

1. Download the 'Legacy MATLAB and Simulink Support for Arduino' package from Mathworks website. Extract the compressed folder named 'ArduinoIO'. From ArduinoIO folder, copy folder 'pde' and paste it in C:\Program Files (or Program files x86)\Arduino\libraries. (The path may be different depending on the installation directory of the IDE.) After pasting the pde folder in the correct location, open Arduino IDE. If you have pasted the folder in the correct location, you can find 'pde' by clicking File→Examples→pde. Open the code by proceeding as follows: File→Examples→pde→radioes.

2. Connect the Arduino Uno board to your PC. From Device Manager, note the COM port at which the Arduino Uno board is installed. From Tools menu in the Arduino IDE, select the board as Arduino Uno, followed by the COM Port number noted earlier. Upload 'adioes' code to the Arduino Uno board by pressing 'Upload' button in the IDE.

3. Now copy the entire contents of the extracted ArduinoIO folder to a folder in 'My Documents' (for Windows PC).

4. This GUI application program is developed in the R2014a version of MATLAB. After installing this version of MATLAB in your PC, open the file 'install_arduino.m' from the directory where you have copied the contents of ArduinoIO folder. Now run the install_arduino.m file. The code will correctly install and save the path of the Arduino support package.

5. Open the project's source code file titled 'relay_control_wireless.m'. Edit the line `a = arduino ('COM9')` with the COM port number in your PC where the Arduino Uno board has been installed. When you run